

Extract more parallelism in every generation

	Nehalem	Sandy Bridge	Haswell	
Out-of-order Window	128	168	192	↑
In-flight Loads	48	64	72	↑
In-flight Stores	32	36	42	↑
Scheduler Entries	36	54	60	↑
Integer Register File	N/A	160	168	↑
FP Register File	N/A	144	168	↑
Allocation Queue	28/thread	28/thread	56	↑

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INTEL DEVELOPER FORUM

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Haswell Buffer Sizes

implementation of this depends entirely on the display manufacturer, which is why we are likely to see less devices supporting this feature than one might expect.

Haswell has opened new horizons for Intel in terms of the performance to power consumption ratio that it has traditionally been able to deliver. While the platform is certainly not ready to move into the domain where Atom currently resides, it is certainly not a replacement for such devices. However, it does pave a way for future platforms which would be able to fit this architecture into mobile phones, thus providing unparalleled performance along with a respectable battery life for such powerful devices.

ARCHITECTURAL IMPROVEMENTS

With the Tick-Tock development model proposed by Intel, Haswell borrows a lot from the previous generation of Ivy Bridge processors and attempts to build upon the basics of that technology while making a lot of changes to the architecture. The front end in Haswell includes some improvements in the branch prediction which improves efficiency in fetching and decoding the instructions. It now also has a single 56-entry instruction decode queue along with a 1.5k micro op cache, which caches decoded micro ops,

effectively reducing the pipeline to 14 stages if a cache hit occurs.

Some of the most important changes have been made at the back end in the execution engine, where Out of Order execution window has been expanded to provide more parallelism for the execution pipeline. Apart from expanding the reorder buffer and the unified reservation station, Haswell adds two more execution ports, each for integer math and store address calculation.

INTEL ADVANCED VECTOR EXTENSIONS 2

The Haswell New Instructions or AVX2 provide enhancements to the Advanced Vector Extensions that came with the Sandy Bridge

processor. Advanced Vector Extensions are the 256-bit instruction set for floating point workloads. AVX offers three-operand non-destructive SSE instruction where the destination register is different and doesn't overwrite the one for source operand. AVX2 also supports Fused Multiply Add (FMA) on two ports i.e. 0 & 1, thus effectively doubling the throughput for floating point operations. Previously, such FMA operation used to take 8 cycles, which has now been reduced to only 5 in Haswell. This has resulted in doubling the bandwidth for the L1 cache as well. The interface between L1 and L2 caches has also been doubled in order to improve performance. Additionally, Gather support has been added, thus enabling sparse vector addressing, so that elements can be loaded from non-contiguous memory locations.

TRANSACTIONAL SYNCHRONIZATION EXTENSIONS

Intel introduced the Transactional Synchronization Extensions, an instruction set separate from the AVX2 which adds transactional memory support so that multi-threaded programs can be executed much more reliably than before. With the help of this instruction set, developers can specify code regions for transactional synchronization, so that complex operations can be performed concurrently in isolation from each other. These transactional features somewhat resemble the similar synchronization paradigm in programming and databases. Haswell's transaction support comes in the form of two parts i.e. Hardware Lock Elision and Restricted Transactional Memory.

Hardware Lock Elision is an approach which is legacy compatible and is compatible with the normal lock-based programming model. HLE is supported even on hardware which doesn't support TSX, such as the newer K-series high-end processors. It uses the XAQUIRE/XRELEASE instructions to mark a code for synchronized access. If a conflict occurs, then the code is executed with traditional locking mechanism.

Restricted Transactional Memory is the new interface that allows notation of transactional regions in a manner

Metric	Nehalem	Sandy Bridge	Haswell
L1 Instruction Cache	32K, 4-way	32K, 8-way	32K, 8-way
L1 Data Cache	32K, 8-way	32K, 8-way	32K, 8-way
Fastest Load-to-use	4 cycles	4 cycles	4 cycles
Load bandwidth	16 Bytes/cycle	32 Bytes/cycle (banked)	64 Bytes/cycle
Store bandwidth	16 Bytes/cycle	16 Bytes/cycle	32 Bytes/cycle
L2 Unified Cache	256K, 8-way	256K, 8-way	256K, 8-way
Fastest load-to-use	10 cycles	11 cycles	11 cycles
Bandwidth to L1	32 Bytes/cycle	32 Bytes/cycle	64 Bytes/cycle
L1 Instruction TLB	4K: 128, 4-way 2M/4M: 7/thread	4K: 128, 4-way 2M/4M: 8/thread	4K: 128, 4-way 2M/4M: 8/thread
L1 Data TLB	4K: 64, 4-way 2M/4M: 32, 4-way 1G: fractured	4K: 64, 4-way 2M/4M: 32, 4-way 1G: 4, 4-way	4K: 64, 4-way 2M/4M: 32, 4-way 1G: 4, 4-way
L2 Unified TLB	4K: 512, 4-way	4K: 512, 4-way	4K+2M shared: 1024, 8-way

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The 2X cache bandwidth in Haswell